

★ PRIMER · UNIT 11

Before the Lecture: Why Two Is Safer Than One

Unit 11 · Portfolio Construction

Watch this first if portfolio variance and correlation are hazy

Twenty minutes of groundwork — how two assets combine, why correlation is the whole story, and what happens when you invert a matrix built mostly of noise.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Portfolio variance is here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 11 Sits

The course builds one pipeline; here is this unit's stone

Unit 10

done

Testing that does not lie to you.

Unit 11

you are here

Building portfolios that survive.

Unit 12

next

Teaching machines to read.

Unit 9 cleaned the covariance matrix. This unit finally spends it.

Where Diversification Comes From

Two assets that do not move together have a combined risk lower than either one alone. That gap is the free lunch.

The return

- A weighted average.
- Nothing surprising.
- Half and half gives the middle.

The risk

- Not a weighted average.
- Correlation pulls it down.
- This is the whole gain.

Return is linear in the weights. Risk is not, and that asymmetry is the entire business.

The Words the Lecture Assumes

Five terms, and every formula in the unit opens up

Word	What it means	Here
Weight	share of the money in an asset	0.5 means half
Correlation	co-movement, from -1 to $+1$	the diversification dial
Covariance	correlation times both volatilities	what the maths actually uses
Variance	volatility squared	what portfolios add up
$1/N$	equal weight in everything	the benchmark that keeps winning

Correlation is scale-free and readable. Covariance is what the formulas need.

What Correlation Buys You

Two assets, each 20% volatility, split half and half

Correlation	Portfolio volatility	What happened
+1.0	20.0%	no gain at all
+0.5	17.3%	a useful reduction
0.0	14.1%	the classic free lunch
-1.0	0.0%	risk cancels entirely

Nothing was given up in return, and risk fell by a third. That is why it is called free.

Why the Elegant Answer Is Fragile

Markowitz needs expected returns. Change one forecast by half a percentage point and the portfolio can invert.

What it needs

- Expected return per asset.
- A full covariance matrix.
- Then it inverts that matrix.

What goes wrong

- Forecasts are nearly useless.
- The matrix is mostly noise.
- Inversion amplifies both.

Noisy in, crazy out. The lecture shows the flip on a half-point change, and it is startling.

Portfolio Variance by Hand

Two assets, equal weights, 20% volatility each, correlation 0.5

Term	Formula piece	Value
Asset A	$0.5^2 \times 0.20^2$	0.0100
Asset B	$0.5^2 \times 0.20^2$	0.0100
Cross term	$2 \times 0.5 \times 0.5 \times 0.5 \times 0.20^2$	0.0100
Variance	sum of the three	0.0300
Volatility	square root	17.3%

Both assets are 20% on their own. Together they are 17.3%, and nothing was sacrificed.

Reading That Cross Term

One term carries the entire idea of diversification

If correlation were 1

- The cross term doubles.
- Variance is 0.04.
- Volatility back to 20%.

If correlation were 0

- The cross term vanishes.
- Variance is 0.02.
- Volatility falls to 14.1%.

Diversification is not about counting assets — it is about how they move together

Respect the Humble 1/N

Equal weights beat clever optimisers more often than anyone expects

Why it should lose

- It ignores every forecast.
- It ignores every correlation.
- It is almost embarrassing.

Why it often wins

- It has nothing to estimate.
- So nothing can be estimated wrong.
- No matrix, no inversion.

Any method you propose must beat 1/N out of sample. Most published ones do not.

How HRP Avoids the Inversion

Three moves, and none of them inverts anything

1 · Cluster

by similarity

Group assets that move together, using unit 9's tools.

2 · Order

the matrix

Rearrange so similar assets sit next to each other.

3 · Split

risk downward

Divide capital between groups, then within them, recursively.

No matrix inversion anywhere. That is precisely why it survives a noisy covariance matrix.

Four Ways to Set Weights

The lecture tests all four on the same data

Method	Needs forecasts?	Inverts a matrix?
1/N	no	no
Inverse volatility	no	no
Markowitz	yes	yes
HRP / NCO	no	no

The two columns explain the results. Every fragility traces back to a yes.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
w	w	the vector of portfolio weights
Σ	sigma	the covariance matrix
ρ	rho	correlation between two assets
σ_p	sigma p	the portfolio's own volatility

Portfolio variance is w -transpose Σ w . The hand calculation you did is that formula.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>returns.cov()</code>	the covariance matrix
<code>returns.corr()</code>	the readable version of the same thing
<code>w @ cov @ w</code>	portfolio variance, in one line
<code>np.linalg.inv(cov)</code>	what Markowitz needs — and what breaks
<code>np.ones(n) / n</code>	the 1/N benchmark you must beat

Print the condition number of your covariance matrix before inverting it. It is sobering.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 12

1

"Holding fifty stocks means I am diversified"

Fifty stocks that all move together is one position. Correlation decides, not the count.

2

"Markowitz is optimal, so it is best"

It is optimal given its inputs. Its inputs are estimates, and it amplifies their errors.

3

"1/N is a naive straw man"

It is the benchmark that most published methods quietly fail to beat out of sample.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1 Two assets, 20% vol, correlation 0. Portfolio vol at 50/50?

14.1%

2 Which step makes Markowitz fragile?

inversion

3 What must any new method beat out of sample?

1/N

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It prices the free lunch

Portfolio risk by hand, and how correlation — not the number of names — creates the gain.

2

It breaks Markowitz on stage

A half-point change in one forecast, and the optimal portfolio flips completely.

3

It ends in code

The problem set breaks Markowitz, then builds HRP on a denoised matrix from unit 9.

Next: Unit 11 lecture — Building Portfolios That Don't Break